Points 6 Published

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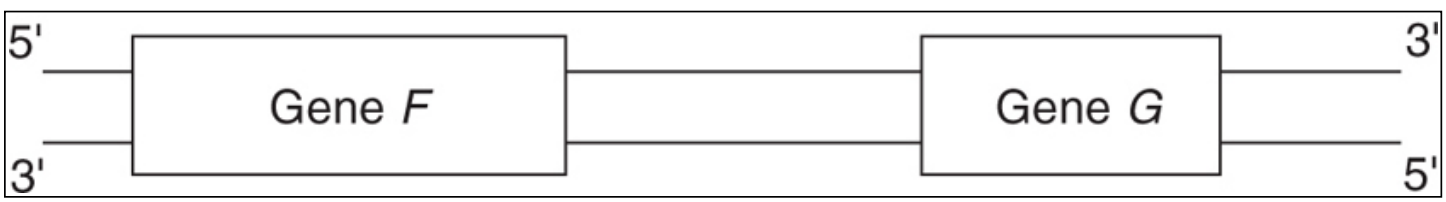
Question

Here is a codon table, for your reference.

1st letter	Second Letter: U		Second Letter: C		Second Letter: A		Second Letter: G		3rd letter
U	UUU	Phe	UCU	Ser	UAU	Tyr	UGU	Cys	U
	UUC		UCC		UAC		UGC		C
	UUA	Leu	UCA		UAA	Stop	UGA	Stop	A
	UUG		UCG		UAG	Stop	UGG	Trp	G
C	CUU	Leu	CCU	Pro	CAU	His	CGU	Arg	U
	CUC		CCC		CAC		CGC		C
	CUA		CCA		CAA	Gln	CGA		A
	CUG		CCG		CAG		CGG		G
A	AUU	Ile	ACU	Thr	AAU	Asn	AGU	Ser	U
	AUC		ACC		AAC		AGC		C
	AUA		ACA		AAA	Lys	AGA	Arg	A
	AUG	Met	ACG		AAG		AGG		G
G	GUU	Val	GCU	Ala	GAU	Asp	GGU	Gly	U
	GUC		GCC		GAC		GGC		C
	GUA		GCA		GAA	Glu	GGA		A
	GUG		GCG		GAG		GGG		G

Question 1 pts

The coding sequence for gene F is read from left to right on the following figure. The coding sequence for gene G is read from right to left. Which strand of DNA (top or bottom) serves as the template for transcription of Gene F?



Top

Bottom

Question 1 pts

Which strand of DNA (top or bottom) serves as the template for transcription of Gene G?

Top

Bottom

Question 1 pts

The sequence of a complete eukaryotic gene encoding the small protein Met Tyr Arg Gly Ala is shown here.

```
5′ CCCCTATGCCCCCCTGGGGGAGGATCAAAACACTTACCTGTACATGGC 3′
3′ GGGGATACGGGGGGACCCCCTCCTAGTTTTGTGAATGGACATGTACCG 5′
```

What is the sequence of the nucleotides in the processed mRNA molecule for this gene? Indicate the 5' and 3' polarity of this mRNA (e.g. 5'-AGCGAGACG-3'). Use uppercase letters and remember that RNA uses uracil as a base. Hint: Most eukaryotic genes have a 5' UTR, introns, and a 3' UTR—not just the protein coding sequence.

5'-GCCAUGUACAGGGGGGCA

Question 1 pts

A single base mutation in the gene results in the synthesis of the peptide Met Tyr Ser. What is the most likely sequence of nucleotides making up the mRNA produced by this mutant gene? Indicate the 5' and 3' polarity of this mRNA. Use uppercase letters. There are multiple correct answers.

5'-GCCAUGUACAGUUAAGUGU

5'-GCCAUGUACAGCUAAGUGU

Question

From two true-breeding lines, you get a dihybrid plant that is heterozygous for a gene that makes red pigment for flower color (*Aa*) and heterozygous at the $tRNA^{Tyr}$ locus (B^TB^N). The *a* allele contains a nonsense mutation (creates a stop codon) that cannot produce the red pigment. The *A* allele shows incomplete dominance, where only one functional copy will produce pink flowers. The B^N allele is a nonsense-suppressor tRNA that can insert a tyrosine at a stop codon, thus allowing for a fully functional peptide for allele *a*. Assume you also have another copy of the wild-type version of the $tRNA^{Tyr}$ gene at a different locus and that the suppressor tRNA is inefficiently used at normal stop codons. Also assume that one copy of the suppressor tRNA can create fully functional peptides of both copies of *a,* if the A locus is homozygous recessive (*aa*).

Question 1 pts

What is the color of the flowers of the dihybrid plant?

red
pink
white

Question 1 pts

If you were to self the dihybrid, what is the expected phenotypic ratio of red:pink:white progeny? Hint: Think about what genotypes can give rise to each phenotype and use product and sum rules to figure out final phenotypic ratios. You can also use a punnett square.

9:3:3:1
12:3:1

9:7
13:2:1
13:3
15:1

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